

# Cem Ozcelik

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## PROFESSIONAL SUMMARY

Senior Data Analyst / Data Scientist with an Industrial Engineering background and nearly 5 years of progressive experience translating large complex datasets into actionable business insights. Deep expertise in advanced SQL on PostgreSQL/MSSQL/Oracle, statistical analysis, KPI tracking and dashboards, A/B testing, and ETL pipelines -- across retail, e-commerce-adjacent, and manufacturing contexts. Sample impact: Route-Level Profitability Analysis using advanced SQL delivered a 3% improvement in logistics efficiency; SKU-level Association Rule Mining drove cross-sell revenue growth; predictive supply chain analytics sustained an 8-hour weekly reduction in operational overtime. Comfortable working cross-functionally with product, BI, and engineering teams. Big-data fluency on PySpark/Databricks; basic-to-advanced ML knowledge (XGBoost, scikit-learn, recommendation systems). Demo: [github.com/ccemozclk/amazon-ai-search-engine](https://github.com/ccemozclk/amazon-ai-search-engine) | [github.com/ccemozclk/Spark-Recommendation-Engine](https://github.com/ccemozclk/Spark-Recommendation-Engine)

## CORE COMPETENCIES

**Skills:** Advanced SQL on Large Complex Datasets, KPI Tracking & Dashboards, Merchant / Customer Performance Analytics, A/B Testing & Funnel Analysis, Statistical Analysis, Cross-functional Stakeholder Communication, Data Quality & ETL, Data Modeling

**Tools:** SQL (Advanced), Python, PySpark, Databricks, Streamlit, Django, FastAPI, Docker, Git, scikit-learn, XGBoost, LightGBM, Qdrant (Vector DB), Redis

## WORK EXPERIENCE

### Anadolu Pet Akvaryum & Effeffe

Jan 2026 -- Present

#### Senior Database Analyst | Maltepe, Istanbul

- Acted as **strategic data partner for business and product teams**, translating SKU-level Association Rule Mining outputs into cross-sell actions that drove continuous revenue growth.
- Directed predictive supply chain analytics using advanced SQL and Python, sustaining an **8-hour weekly reduction in operational overtime** and surfacing actionable KPIs for operations leadership.
- Led cross-functional data initiatives across BI, engineering, and operations -- defining metrics, ensuring data quality, and translating complex analyses into clear insights for non-technical stakeholders.
- Scaled the company's Django data application into a centralized hub serving versioned REST APIs for analytics and ML models.
- Tech:** Advanced SQL, Python, Django, PostgreSQL, REST APIs, ETL, Docker

### Anadolu Pet Akvaryum & Effeffe

Sept 2024 -- Dec 2025

#### IT & Data Analytics Specialist | Maltepe, Istanbul

- Conducted **Route-Level Profitability Analysis with advanced SQL**, defining cost-to-profit ratios that delivered a **3% improvement in overall logistics efficiency** -- direct analog to merchant performance evaluation in marketplace contexts.
- Engineered predictive supply chain models in Python and PL/SQL that **reduced stockouts and improved inventory turnover**.
- Architected a full-stack internal web application (Python/Django, SOLID principles), establishing the foundation for KPI dashboards and reporting.
- Tech:** Advanced SQL, PL/SQL, Oracle, Python, Django, full-stack deployment

### Birinci Automotive

Jan 2022 -- Sept 2024

#### Mid-Level Data Analyst | Cayirova, Kocaeli

- Engineered robust ETL pipelines in MSSQL and Python to ingest and process high-volume manufacturing and machine sensor data, replacing manual reporting with reliable analytical infrastructure.
- Developed an **unsupervised clustering framework on behavioral time-series data** for anomaly detection -- the same statistical primitives that drive customer segmentation and KPI anomaly alerting.
- Bridged raw production data and operational strategy, delivering a **15% weekly increase in critical machine availability** and **15% OEE improvement**.
- Tech:** Advanced SQL (MSSQL), Python, ETL, unsupervised clustering, time-series analysis

Junior Data Analyst | Cayirova, Kocaeli

- Conducted **Exploratory Data Analysis (EDA)** and **applied statistical methods** to workforce datasets, uncovering hidden patterns to optimize HR policies.
- Designed automated data workflows and ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

E-Commerce AI Search Engine Semantic Search & A/B Testing

Production-ready microservices architecture with **dual-engine A/B testing**: SentenceTransformers (all-MiniLM-L6-v2) vs custom-trained Word2Vec on **800K+ product descriptions** -- the same A/B framework used for marketplace search and ranking experiments. **Item2Vec collaborative filtering** for "Frequently Bought Together" recommendations on hundreds of thousands of user sessions. Sub-millisecond cached responses via Redis; PySpark distributed ETL for massive JSONL ingestion.

PyTorch, FastAPI, Streamlit, Qdrant (Vector DB), Redis, PySpark, Docker | [github.com/ccemozclk/amazon-ai-search-engine](https://github.com/ccemozclk/amazon-ai-search-engine)

Scalable Recommendation Engine PySpark & Customer Segmentation

End-to-end personalized recommendation system on Apache Spark/Databricks. **RFM analysis + K-Means** for behavioral customer segmentation into 5 strategic tiers (Champions, At-Risk, etc.) -- direct analog to merchant tiering and customer KPI segmentation in e-commerce. **ALS collaborative filtering** with implicit feedback and cold-start handling. Validated across hundreds of thousands of user sessions at distributed scale.

PySpark, Apache Spark, Databricks, K-Means, ALS, Streamlit | [github.com/ccemozclk/Spark-Recommendation-Engine](https://github.com/ccemozclk/Spark-Recommendation-Engine)

Retail Demand Forecasting Dashboards & KPI Translation

~**12.83% RMSPE** on 6-week daily sales across 1,115 Rossmann stores with time-based validation (no leakage). Streamlit dashboard with **dynamic confidence intervals** translating ML metrics into business terms (margin of error in EUR per store/day). Surfaced actionable insight: B-type stores yield lower promotion ROI vs A/C, enabling marketing budget reallocation. Object-oriented training pipeline + FastAPI REST inference.

Python, LightGBM, XGBoost, Optuna, FastAPI, Streamlit, Docker | [github.com/ccemozclk/Rossmann-Demand-Forecasting](https://github.com/ccemozclk/Rossmann-Demand-Forecasting)

EDUCATION

B.Sc. Industrial Engineering — Kirikkale University

Sept 2016 -- Feb 2021

Quantitative degree with rigorous foundation in statistical analysis, operations research, and optimization. Designed MILP models with IBM ILOG CPLEX. Applied AHP and TOPSIS to multi-criteria decision problems -- direct prerequisite for KPI design and merchant performance evaluation in marketplace settings.

CERTIFICATIONS

Associate Data Scientist Career Track (incl. PySpark, Recommendation Engines) — DataCamp

Jan -- May 2024

Data Science & Machine Learning Bootcamp (6 months, 10+ projects) — Miuul

Mar -- Sept 2022

SKILLS

**Languages:** Python (Advanced), SQL (Advanced), Bash

**Analytics & Statistical:** EDA, A/B Testing & Experiment Design, Statistical Analysis, KPI Modeling, Funnel Analysis, Data Quality, Customer Segmentation (RFM, K-Means), Recommendation Systems (ALS, Item2Vec)

**Big Data:** PySpark, Apache Spark, Databricks

**BI & Dashboards:** Streamlit, Django (interactive EDA platforms), FastAPI -- transferable to Looker/Qlik/BigQuery

**Databases:** PostgreSQL, MSSQL, Oracle, ETL Pipelines, Redis, Qdrant (Vector DB)

**ML:** scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, h2o.ai, CNNs, RNNs

**MLOps:** Docker, FastAPI, Django, Flask, REST APIs, Git, Linux, AWS (EC2, S3)

**Advanced:** Cost-Function Optimization, MILP, AHP/TOPSIS, OOP, System Architecture     **Languages:** Turkish (Native), English (Fluent)